

Model dossier — VLC_LN

Status: reverse run failed. The blank-slate reconstruction aborted on an agent transport error. Infrastructure, not physics.

This dossier is not a completed reverse-check review package. It collects what exists for this model so you can read the physics without opening the source files. Where a full review package does exist for a model, it is the REVIEW.pdf in that model's directory.

item	value
model	VLC_LN
chain status	passed the full chain
Lagrangian source	repair2/final.fr (re-validated)
Lagrangian terms found	15

Verbatim Lagrangian terms

Quoted unmodified from the .fr. These are the terms any reconstruction would have to account for.

PiMat (=)

```
{{pi0/Sqrt[2] + eta/Sqrt[6] + etaP/Sqrt[3], pip, Kp}, {pipbar, -pi0/Sqrt[2] + eta/Sqrt[6] +  
↪ etaP/Sqrt[3], K0}, {Kpbar, K0bar, -2 eta/Sqrt[6] + etaP/Sqrt[3]}}
```

Umat (:=)

```
MatrixExp[I Sqrt[2] PiMat/fpi]
```

Mmat (=)

```
{{mL, 0, y PhiSM[1]}, {0, mL, y PhiSM[2]}, {Conjugate[ytil] PhiSMBar[1], Conjugate[ytil]  
↪ PhiSMBar[2], mN}}
```

LUVVLC (:=)

```
Block[{mu, sp, aa, kin, yuk},  
  kin = I L0bar[sp, aa].Ga[mu].DC[L0[sp, aa], mu] +  
        I Lmbar[sp, aa].Ga[mu].DC[Lm[sp, aa], mu] +  
        I Nvlcbar[sp, aa].Ga[mu].DC[Nvlc[sp, aa], mu];  
  yuk = y PhiSM[1] L0bar[sp, aa].ProjM.Nvlc[sp, aa] +  
        y PhiSM[2] Lmbar[sp, aa].ProjM.Nvlc[sp, aa] +  
        ytil PhiSMBar[1] L0bar[sp, aa].ProjP.Nvlc[sp, aa] +  
        ytil PhiSMBar[2] Lmbar[sp, aa].ProjP.Nvlc[sp, aa];  
  1/2 (kin + HC[kin]) -  
  mL (L0bar[sp, aa].L0[sp, aa] + Lmbar[sp, aa].Lm[sp, aa]) -  
  mN Nvlcbar[sp, aa].Nvlc[sp, aa] +  
  yuk + HC[yuk]  
]
```

LChiralFull (:=)

```
Block[{mu, i}, fpi^2/4 Tr[DC[Umat, mu].DC[HC[Umat], mu]] + (grho fpi^3 Tr[Mmat.Umat] + HC[grho  
↪ fpi^3 Tr[Mmat.Umat]]) + fpi^2/16 (mEtaP^2/3) (Log[Det[Umat]] - Log[Det[HC[Umat]]])^2 + 3  
↪ gw^2 grho^2 fpi^4/(2 (4 Pi)^2) Sum[Tr[Umat.Ta[i].HC[Umat].Ta[i]], {i, 1, 3}]
```

LKinCompositeScalars (:=)

```
Block[{mu}, 1/2 del[pi0,mu] del[pi0,mu] + del[pipbar,mu] del[pip,mu] + 1/2 del[eta,mu]
  ↳ del[eta,mu] + 1/2 del[etaP,mu] del[etaP,mu] - mEtaP^2/2 etaP^2]
```

LMExpanded (:=)

```
-mK2^2 (Kpbar Kp + K0bar K0) - mPi3^2/2 (pi0^2 + 2 pip pipbar) - mEta^2/2 eta^2 + (-I
  ↳ Sqrt[2] grho fpi^2 Bcoup (Kpbar PhiSM[1] + K0bar PhiSM[2]) - grho/Sqrt[2] Acoup fpi
  ↳ (Sum[(Kpbar PauliSigma[a,1,jj] + K0bar PauliSigma[a,2,jj]) PhiSM[jj] PiTriplet[a],
  ↳ {a,1,3}, {jj,1,2}] - eta (Kpbar PhiSM[1] + K0bar PhiSM[2])/Sqrt[3]) + HC[-I Sqrt[2] grho
  ↳ fpi^2 Bcoup (Kpbar PhiSM[1] + K0bar PhiSM[2]) - grho/Sqrt[2] Acoup fpi (Sum[(Kpbar
  ↳ PauliSigma[a,1,jj] + K0bar PauliSigma[a,2,jj]) PhiSM[jj] PiTriplet[a], {a,1,3},
  ↳ {jj,1,2}] - eta (Kpbar PhiSM[1] + K0bar PhiSM[2])/Sqrt[3]))]
```

LKinTriplet (:=)

```
(I del[pipbar, mu] gw pi0 WiPhys[mu, 1])/(2 Sqrt[2]) - (I del[pip, mu] gw pi0 WiPhys[mu,
  ↳ 1])/(2 Sqrt[2]) - (I del[pi0, mu] gw pipbar WiPhys[mu, 1])/(2 Sqrt[2]) + (I del[pi0, mu]
  ↳ gw pip WiPhys[mu, 1])/(2 Sqrt[2]) + (del[pipbar, mu] gw pi0 WiPhys[mu, 2])/(2 Sqrt[2]) +
  ↳ (del[pip, mu] gw pi0 WiPhys[mu, 2])/(2 Sqrt[2]) - (del[pi0, mu] gw pipbar WiPhys[mu,
  ↳ 2])/(2 Sqrt[2]) - (del[pi0, mu] gw pip WiPhys[mu, 2])/(2 Sqrt[2]) + 1/2 I del[pip, mu] gw
  ↳ pipbar WiPhys[mu, 3] - 1/2 I del[pipbar, mu] gw pip WiPhys[mu, 3]
```

LKinK2 (:=)

```
Block[{mu}, DK2Bar[1, mu] DK2[1, mu] + DK2Bar[2, mu] DK2[2, mu]]
```

LAnomalyTriplet (:=)

```
Block[{mu,nu,rho,sig}, NHC g1 gw/(32 Pi^2 fpi) Eps[mu,nu,rho,sig] BFS[rho,sig] (PiTriplet[1]
  ↳ SU2FS[mu,nu,1] + PiTriplet[2] SU2FS[mu,nu,2] + PiTriplet[3] SU2FS[mu,nu,3])]
```

LAnomalyEta (:=)

```
Block[{mu,nu,rho,sig}, -NHC eta/(32 Sqrt[3] Pi^2 fpi) Eps[mu,nu,rho,sig] (gw^2
  ↳ (SU2FS[mu,nu,1] SU2FS[rho,sig,1] + SU2FS[mu,nu,2] SU2FS[rho,sig,2] + SU2FS[mu,nu,3]
  ↳ SU2FS[rho,sig,3]) + g1^2 BFS[mu,nu] BFS[rho,sig])]
```

LEDm (:=)

```
Block[{mu,nu,rho,sig,a}, -mPi3^2/2 Sum[PiTriplet[a]^2,{a,1,3}] - mEta^2/2 eta^2 + 4 Im[y
  ↳ ytil] grho^2 fpi^3/mK2^2 (Sum[PhiSMBar[i] PauliSigma[a,i,j] PhiSM[j]
  ↳ PiTriplet[a],{a,1,3},{i,1,2},{j,1,2}] - eta Sum[PhiSMBar[i] PhiSM[i],{i,1,2}]/Sqrt[3]) +
  ↳ NHC g1 gw/(32 Pi^2 fpi) Eps[mu,nu,rho,sig] BFS[rho,sig] (PiTriplet[1] SU2FS[mu,nu,1] +
  ↳ PiTriplet[2] SU2FS[mu,nu,2] + PiTriplet[3] SU2FS[mu,nu,3]) - NHC eta/(32 Sqrt[3] Pi^2
  ↳ fpi) Eps[mu,nu,rho,sig] (gw^2 (SU2FS[mu,nu,1] SU2FS[rho,sig,1] + SU2FS[mu,nu,2]
  ↳ SU2FS[rho,sig,2] + SU2FS[mu,nu,3] SU2FS[rho,sig,3]) + g1^2 BFS[mu,nu] BFS[rho,sig])]
```

LRhoff (:=)

```
1/Sqrt[2] gV (rhoP[mu] (vlbar.Ga[mu].ProjM.l + ubar.Ga[mu].ProjM.d + cbar.Ga[mu].ProjM.s +
  ↳ tbar.Ga[mu].ProjM.b) + rhoPbar[mu] (lbar.Ga[mu].ProjM.vl + dbar.Ga[mu].ProjM.u +
  ↳ sbar.Ga[mu].ProjM.c + bbar.Ga[mu].ProjM.t)) + 1/2 gV rho0[mu] (vlbar.Ga[mu].ProjM.vl +
  ↳ ubar.Ga[mu].ProjM.u + cbar.Ga[mu].ProjM.c + tbar.Ga[mu].ProjM.t - lbar.Ga[mu].ProjM.l -
  ↳ dbar.Ga[mu].ProjM.d - sbar.Ga[mu].ProjM.s - bbar.Ga[mu].ProjM.b)
```

```
LRhoTriplet (:=)
```

```
grho 1/2 I del[pip, mu] pipbar rho0[mu] - grho 1/2 I del[pipbar, mu] pip rho0[mu] - grho 1/2 I del[pip, mu] pi0 rho0bar[mu] + grho 1/2 I del[pi0, mu] pip rho0bar[mu] + grho 1/2 I del[pipbar, mu] pi0 rho0[mu] - grho 1/2 I del[pi0, mu] pipbar rho0[mu]
```

```
LTot (:=)
```

```
LUVVLC + LKinCompositeScalars + LMExpanded + LKinTriplet + LKinK2 + LAnomalyTriplet + LAnomalyEta + LRhff + LRhoTriplet
```

The last failure the loop worked on

This is the validation report handed to the repair agent at the start of `repair2/round2` — the problem it was asked to fix. The model **passed** after this round; the report below is the state that was repaired, not the final one.

Validation report — model.fr FAILED the tool chain

- FeynRules UFO compile: OK
- Hermiticity check: True
- Kinetic-terms check: True
- Mass-spectrum check: True
- MadGraph import: False
- Heuristic error tags: mg5_import_fail, mg5_python_traceback

FeynRules check hermiticity output

Checking for hermiticity by calculating the Feynman rules contained in L-HC[L].
If the lagrangian is hermitian, then the number of vertices should be zero.
Starting Feynman rule calculation.
Expanding the Lagrangian...
Expanding the indices over 8 cores
Collecting the different structures that enter the vertex.
No vertices found.
0 vertices obtained.
The lagrangian is hermitian.

FeynRules check kinetic_terms output

Neglecting all terms with more than 2 particles.
All kinetic terms are diagonal.

FeynRules check mass_spectrum output

Neglecting all terms with more than 2 particles.
Non diagonal mass term found: $(\text{Acoup} \cdot \eta \cdot f_{\pi} \cdot g_{\rho} \cdot K_0 \text{bar} \cdot \text{vev}) / (2 \cdot \text{Sqrt}[3])$
Non diagonal mass term found: $(\text{Acoup} \cdot f_{\pi} \cdot g_{\rho} \cdot K_0 \text{bar} \cdot \pi_0 \cdot \text{vev}) / 2$
Non diagonal mass term found: $-((\text{Acoup} \cdot f_{\pi} \cdot g_{\rho} \cdot K_{\rho} \text{bar} \cdot \pi \cdot \text{vev}) / \text{Sqrt}[2])$
Non diagonal mass term found: $(\eta \cdot f_{\pi} \cdot g_{\rho} \cdot K_0 \cdot \text{vev} \cdot \text{Conjugate}[\text{Acoup}]) / (2 \cdot \text{Sqrt}[3])$
Non diagonal mass term found: $(f_{\pi} \cdot g_{\rho} \cdot K_0 \cdot \pi_0 \cdot \text{vev} \cdot \text{Conjugate}[\text{Acoup}]) / 2$
Non diagonal mass term found: $-((f_{\pi} \cdot g_{\rho} \cdot K_{\rho} \cdot \pi \cdot \text{vev} \cdot \text{Conjugate}[\text{Acoup}]) / \text{Sqrt}[2])$
Warning: not numerical value encountered. Unable to decide whether mass term is diagonal
Non diagonal mass term found: $(\text{vev} \cdot y \cdot L_{\text{bar}}[\text{Index}[\text{Spin}, \text{sp}], \text{Index}[\text{HCIndex}, \text{aa}]] \cdot \text{ProjM} \cdot \text{Nvlc}[\text{Index}[\text{Spin}, \text{sp}], \text{Index}[\text{HCIndex}, \text{aa}]] \cdot \text{ProjM})$
Warning: not numerical value encountered. Unable to decide whether mass term is diagonal
Non diagonal mass term found: $(\text{vev} \cdot \text{Conjugate}[y] \cdot \text{Nvlcbar}[\text{Index}[\text{Spin}, \text{r}\$3285], \text{Index}[\text{HCIndex}, \text{aa}]] \cdot \text{ProjM})$

FeynRules / Wolfram Engine output (tail)

```
rentz, Ext[1]]] . Lm[Index[Spin, SP$1], Index[HCIndex, 2]] + Lmbar[Index[Spin, SP$1], 3] . Ga[Index[Loren
```

```
Symbol::symname: The string "ISUMObjectaI7[Index[Lorentz, Ext[1]]]L0bar[Index[Spin, SP$1], 1] . Ga[0] . G
```

```
General::stop: Further output of Symbol::symname will be suppressed during this calculation.
```

```
From kernel 5 (Local):
```

```
Table::iterb: Iterator {Index[Lorentz, Ext[1]], IndexRange[Index[Lorentz]]} does not have appropriate bou
```

```
From kernel 6 (Local):
```

```
Table::iterb: Iterator {Index[Lorentz, Ext[1]], IndexRange[Index[Lorentz]]} does not have appropriate bou
```

```
From kernel 7 (Local):
```

```
Table::iterb: Iterator {Index[Lorentz, Ext[1]], IndexRange[Index[Lorentz]]} does not have appropriate bou
```

```
From kernel 8 (Local):
```

```
Table::iterb: Iterator {Index[Lorentz, Ext[1]], IndexRange[Index[Lorentz]]} does not have appropriate bou
```

```
From kernel 5 (Local):
```

```
Table::iterb: Iterator {Index[Lorentz, Ext[1]], IndexRange[Index[Lorentz]]} does not have appropriate bou
```

```
General::stop: Further output of Table::iterb will be suppressed during this calculation.
```

```
Flavor expansion of the vertices distributed over 8 cores: Dynamic[FR$Count1] / 176
```

```
- Saved vertices in InterfaceRun[ 1 ].
```

```
QN::NonConserv: Warning: non quantum number conserving vertex encountered!
```

```
Quantum number Y not conserved in vertex {eta, G0, K0bar}.
```

```
QN::NonConserv: Warning: non quantum number conserving vertex encountered!
```

```
Quantum number Y not conserved in vertex {eta, H, K0bar}.
```

```
QN::NonConserv: Warning: non quantum number conserving vertex encountered!
```

```
General::stop: Further output of QN::NonConserv will be suppressed during this calculation.
```

```
Quantum number Q not conserved in vertex {eta, GP, Kpbar}.
```

```
Quantum number Y not conserved in vertex {eta, GP, Kpbar}.
```

```
Quantum number Y not conserved in vertex {G0, K0bar, pi0}.
```

```
Quantum number Y not conserved in vertex {H, K0bar, pi0}.
```

```
Quantum number Q not conserved in vertex {GP, Kpbar, pi0}.
```

```
Quantum number Y not conserved in vertex {GP, Kpbar, pi0}.
```

```
Quantum number Q not conserved in vertex {G0, Kpbar, pip}.
```

```
Quantum number Y not conserved in vertex {G0, Kpbar, pip}.
```

```
Quantum number Q not conserved in vertex {H, Kpbar, pip}.
```

```
Quantum number Y not conserved in vertex {H, Kpbar, pip}.
```

```
Quantum number Y not conserved in vertex {GP, K0bar, pipbar}.
```

```
Quantum number Y not conserved in vertex {eta, G0, K0}.
```

```
Quantum number Y not conserved in vertex {eta, H, K0}.
```

```
Quantum number Q not conserved in vertex {eta, GPbar, Kp}.
```

```
Quantum number Y not conserved in vertex {eta, GPbar, Kp}.
```

```
Quantum number Y not conserved in vertex {G0, K0, pi0}.
```

```
Quantum number Y not conserved in vertex {H, K0, pi0}.
```

```
Quantum number Q not conserved in vertex {GPbar, Kp, pi0}.
```

```
Quantum number Y not conserved in vertex {GPbar, Kp, pi0}.
```

```
Quantum number Y not conserved in vertex {GPbar, K0, pip}.
```

```
Quantum number Q not conserved in vertex {G0, Kp, pipbar}.
```

```
Quantum number Y not conserved in vertex {G0, Kp, pipbar}.
```

```
Quantum number Q not conserved in vertex {H, Kp, pipbar}.
```

```
Quantum number Y not conserved in vertex {H, Kp, pipbar}.
```

```
Quantum number Q not conserved in vertex {GP}.
```

Quantum number Q not conserved in vertex {GPbar}.

Quantum number Q not conserved in vertex {A, pip, W}.

Quantum number Q not conserved in vertex {A, pipbar, Wbar}.

Quantum number Q not conserved in vertex {pip, W, Z}.

Quantum number Q not conserved in vertex {pipbar, Wbar, Z}.

Quantum number Q not conserved in vertex {A, K0bar, Kp, W}.

Quantum number Q not conserved in vertex {K0bar, Kp, W}.

Quantum number Q not conserved in vertex {pi0, pip, W}.

Quantum number Q not conserved in vertex {A, A, pip, W}.

Quantum number Q not conserved in vertex {A, pip, W, Z}.

Quantum number Q not conserved in vertex {A, K0, Kpbar, Wbar}.

Quantum number Q not conserved in vertex {K0, Kpbar, Wbar}.

Quantum number Q not conserved in vertex {pi0, pipbar, Wbar}.

Quantum number Q not conserved in vertex {A, A, pipbar, Wbar}.

Quantum number Q not conserved in vertex {A, pipbar, Wbar, Z}.

Quantum number Q not conserved in vertex {K0bar, Kp, W, Z}.

Quantum number Q not conserved in vertex {pip, W, Z, Z}.

Quantum number Q not conserved in vertex {K0, Kpbar, Wbar, Z}.

Quantum number Q not conserved in vertex {pipbar, Wbar, Z, Z}.

Preparing Python output.

- Splitting vertices into building blocks.

Splitting of vertices distributed over 8 kernels.

- Optimizing: Dynamic[PRIVATE`PY\$SplitVertexCounter]/254 .
- Writing files.

Done!

[INFO] Done.

MadGraph import output (tail)

```
aMC/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
File "$MG5_PATH/models/import_ufo.py", line 413, in import_full_model
    ufo_model = ufomodels.load_model(model_path, decay)
File "$MG5_PATH/models/__init__.py", line 101, in load_model
    raise UFOError(str(error))
models.UFOError: invalid syntax (object_library.py, line 268)
Command "import /private/tmp/repair_bench2/VLC_LN/round1/mg5run/mg5_import.txt" interrupted in sub-command
"import model /tmp/repair_bench2/VLC_LN/round1/UFO" with error:
UFOError : invalid syntax (object_library.py, line 268)
Please report this bug on https://bugs.launchpad.net/mg5amcnlo
More information is found in 'MG5_debug'.
Please attach this file to your report.
```

MG5_debug (the real MadGraph error)

```
"$MG5_PATH/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
File "$MG5_PATH/models/import_ufo.py", line 415, in import_full_model
    model = ufo2mg5_converter.load_model()
File "$MG5_PATH/models/import_ufo.py", line 547, in load_model
    raise InvalidModel("name %s define multiple time. Please correct the UFO model!" \
                        % (param.name))
models.import_ufo.InvalidModel: name WL define multiple time. Please correct the UFO model!
```

During handling of the above exception, another exception occurred:

```

Traceback (most recent call last):
  File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1570, in onecmd
    return self.onecmd_orig(line, **opt)
    ~~~~~^~~~~~
  File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig
    return func(arg, **opt)
  File "$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import
    self.cmd.do_import(self, *args, **opts)
    ~~~~~^~~~~~
  File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5893, in do_import
    self.import_command_file(args[1])
    ~~~~~^~~~~~
  File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1718, in import_command_file
    self.exec_cmd(line, precmd=True)
    ~~~~~^~~~~~
  File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1599, in exec_cmd
    stop = Cmd.onecmd_orig(current_interface, line, **opt)
  File "$MG5_PATH/madgraph/interface/extended_cmd.py", line 1519, in onecmd_orig
    return func(arg, **opt)
  File "$MG5_PATH/madgraph/interface/master_interface.py", line 281, in do_import
    self.cmd.do_import(self, *args, **opts)
    ~~~~~^~~~~~
  File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5835, in do_import
    raise err
  File "$MG5_PATH/madgraph/interface/madgraph_interface.py", line 5819, in do_import
    self._curr_model = import_ufo.import_model(args[1], prefix=prefix,
    ~~~~~^~~~~~
    complex_mass_scheme=self.options['complex_mass_scheme'],
    ~~~~~^~~~~~
    options=options)
    ~~~~~^~~~~~
  File "$MG5_PATH/models/import_ufo.py", line 251, in import_model
    model = import_full_model(model_path, decay, prefix)
  File "$MG5_PATH/models/import_ufo.py", line 413, in import_full_model
    ufo_model = ufomodels.load_model(model_path, decay)
  File "$MG5_PATH/models/__init__.py", line 101, in load_model
    raise UFOError(str(error))
models.UFOError: invalid syntax (object_library.py, line 268)
Fail to write options with error No model currently active, please import a model!

```

Repair history

What the loop tried, and what each attempt measured.

Repair history — previous attempts on this model

Attempt 1

What was changed (the agent's own summary): Updated `model.fr` in place: changed eta-prime's UFO particle name from `eta'` to `etaP` to stop the generated Python string/object-name breakage, and replaced raw `Wi/FS[Wi]/FS[B]` usage in the exported electroweak scalar kinetic and anomaly terms with explicit physical-field component helpers for `W`, `Wbar`, `A`, and `Z`, preventing unevaluated `Wi` symbols from leaking into UFO vertices. `LTot` remains the final top-level assignment and still uses only defined BSM terms; I could not rerun the full validation because `wolframscript` is installed but has no configured Wolfram kernel path.

Validation outcome AFTER that change: status=compiled, hermiticity=True, kinetic=True, mass=True, madgraph_import=False, tags=mg5_import_fail, mg5_python_traceback, compile_seconds=76.2

For reference — the ORIGINAL model's outcome before any repair:

status=compiled, hermiticity=True, kinetic=True, mass=True, madgraph_import=False, tags=mg5_import_fail, ufo_python_syntax_error, mg5_python_traceback, mg5_invalid_cmd, compile_seconds=82.8

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: